

Lab 11 - Median Filter

June 7, 2025

1 Theory

The **median filter** is a non-linear digital filtering technique, often used to remove noise from images or signals. It works by moving a window (usually 3×3) over each pixel in the image, replacing the center pixel with the median value of all the pixels within the window. Unlike linear filters, the median filter preserves edges while effectively reducing impulsive or “salt-and-pepper” noise.

Given an input image I and a window W of size $m \times n$, the median filter output $I'(x, y)$ at position (x, y) is defined as:

$$I'(x, y) = \text{median} \{ I(i, j) \mid (i, j) \in W_{x, y} \}$$

where $W_{x, y}$ is the set of coordinates covered by the window centered at (x, y) .

```
[1]: import numpy as np
      from PIL import Image
      import matplotlib.pyplot as plt
```

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[2]: def median_filter(input_image):
      img = input_image.resize((400,400), Image.Resampling.LANCZOS)

      fig = plt.figure()
      fig.set_figheight(20)
      fig.set_figwidth(20)

      #plotting original image
      fig.add_subplot(1,2,1)
      plt.imshow(img, cmap='gray')
      plt.title('original')

      # convert to numpy array
      numpy_image = np.array(img)
      # array for padding
      array_b = np.zeros((402,402))

      # to pad initial array with zeros in all side
      array_b[1:401,1:401] = numpy_image

      #defining filter
```

```

#filter_array = np.array([[1,1,1],
#                          [1,1,1],
#                          [1,1,1]])

filter_array = np.array([[3,3,3],
                        [3,3,3],
                        [3,3,3]])

#creating empty list
lst = []

for i in range(400):
    for j in range(400):
        #extracting part of array equal to filter size
        array_c = array_b[i:(3+i),j:(3+j)]

        #applying filter
        array_mul = np.multiply(filter_array,array_c)
        array_sum = np.median(array_mul)

        # putting calculated value in list
        lst.append(array_sum)

# resizing lst to shape of original array
final_array = np.resize(lst,(400,400))

final_image = Image.fromarray(final_array)
final_image= final_image.convert("L")

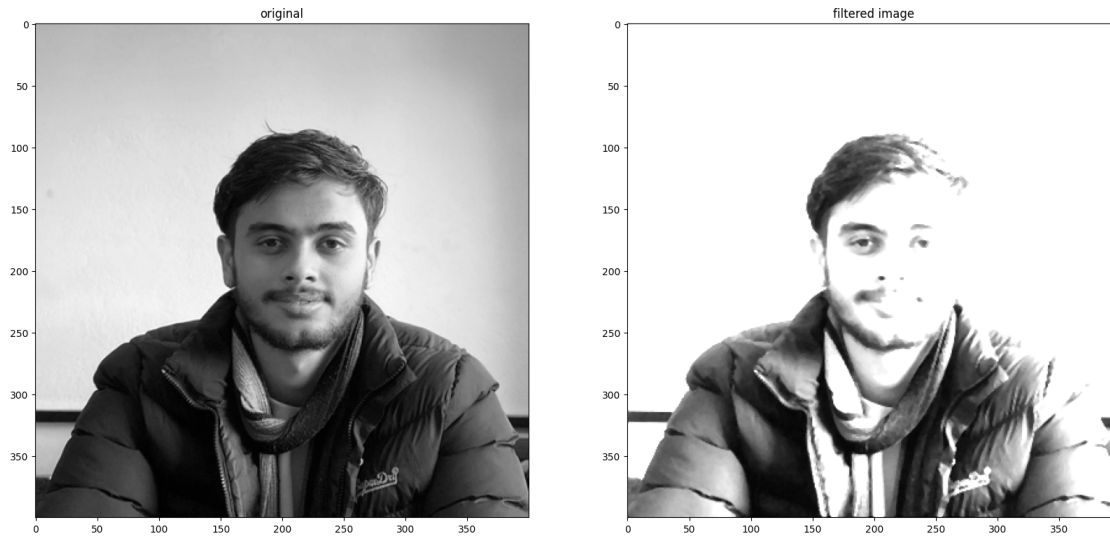
#plotting filtered image
fig.add_subplot(1,2,2)
plt.imshow(final_image, cmap='gray')
plt.title('filtered image')

```

```

[3]: # reading image and converting to gray scale
img = Image.open('img/samir.jpg').convert('L')
# Call filter function
median_filter(img)

```



2 Conclusion

The median filter is highly effective for removing impulsive noise while preserving important image features such as edges. In this lab, the filter was applied to a grayscale image, demonstrating significant noise reduction without blurring the image. This makes the median filter a preferred choice for preprocessing in many image processing applications.